Practice Questions: Units 1

Unit 1: Basic Principle of Counting, Permutations and Combinations

Short Answer Questions

1. How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 if:
   (a) repetition is not allowed,
   (b) repetition is allowed.
2. Find the number of ways in which 4 boys and 3 girls can be seated in a row such that:
   (a) all boys sit together,
   (b) boys and girls sit alternately.
3. How many 4-letter words can be formed from the word **NEPAL**?
4. In how many ways can the letters of the word **MANAGEMENT** be arranged?
5. From a group of 7 men and 5 women, a committee of 4 people is to be formed. In how many ways can this be done if the committee includes:
   (a) exactly 2 women,
   (b) at least one woman.

Long Answer Questions

6. A class has 5 boys and 4 girls. In how many ways can a group of 5 students be formed such that:
   (a) at least 2 girls are included,
   (b) no restriction.
7. Find the number of ways of arranging the letters of the word **PROBABILITY** such that the vowels always come together.
8. How many 5-letter words can be formed from the letters of the word **MATHS** if:
   (a) the word begins with the letter **M**,
   (b) the word does not begin with the letter **M**,
   (c) the word begins with **M** and does not end with **S**.
9. A password consists of 2 English letters followed by 3 digits. How many such passwords can be formed if:
   (a) repetition is allowed,
   (b) repetition is not allowed.
10. Find the number of 5-digit even numbers that can be formed using the digits 1, 2, 3, 4, 5, 6 without repetition.
11. A team of 6 students is to be formed from 8 boys and 7 girls. Find the number of ways to form the team if:
    (a) it contains at least 3 girls,
    (b) it contains more girls than boys.

Unit 1B: Permutations and Combinations

Short Answer Questions

1. Evaluate: ${}^7P_4$ and ${}^7C_4$.
2. If $nP_3 = 120$, find n.
3. Find the number of ways to arrange the letters of the word **NEPAL**.
4. In how many ways can 3 girls be selected from a group of 5?
5. Prove that ${}^nC_r = {}^nC_{n-r}$.
6. Find r if ${}^8C_r = {}^8C_{r+2}$.
7. Find the value of $\frac{{}^8P_3}{{}^8C_3}$.
8. In how many ways can 3 people be chosen from 6?
9. A committee of 3 men and 2 women is to be formed from 5 men and 4 women. In how many ways can this be done?
10. How many 3-digit even numbers can be formed using digits 2, 4, 6, 7, and 8 without repetition?

Long Answer Questions

11. How many 5-digit numbers can be formed using digits 1–5 if:
    (a) repetition is not allowed?
    (b) repetition is allowed?
12. Prove that ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.
13. In how many ways can the letters of the word **SUCCESS** be arranged?
14. From 6 men and 4 women, how many committees of 5 can be formed that include at least 2 women?
15. If ${}^nP_2 = 56$, find n.
16. How many arrangements can be made of the letters of the word **MANAGEMENT** so that all vowels occur together?
17. From a group of 5 boys and 3 girls, how many teams of 4 can be formed with exactly 2 girls?
18. Find the number of circular permutations of 6 boys around a round table.
19. Find the number of 4-digit numbers greater than 5000 that can be formed using digits 2, 3, 5, 6, and 7.
20. A committee of 4 is to be formed from 7 men and 5 women. How many ways are there to form a committee with at most 2 women?